

PLUGGING INTO THE FUTURE

An Exploration of Electricity Consumption Patterns Across India (2019–2020)

Project Documentation

Tools: MySQL · Tableau · Flask · Python

1. Problem Statement

Electricity demand in India varies widely across states and regions, shaped by climate, industrial activity, population density, and — as seen in 2020 — public health emergencies. This project analyzes daily state-wise electricity consumption from January 2019 through December 2020 to uncover regional demand patterns, seasonal trends, and the effect of the COVID-19 lockdown (from 25 March 2020) on power usage, and to present these findings through an interactive Tableau dashboard and data story.

2. Project Overview

The project follows five stages, mirroring the Kanban board used to plan the work:

- Data Collection & Extraction — sourcing and validating the raw dataset
- Data Preparation — storing data in a database, running SQL operations, connecting to Tableau, and shaping fields for analysis
- Data Visualization — building individual chart views in Tableau
- Dashboard & Story — combining views into an interactive, responsive dashboard and a narrative storyboard
- Performance Testing & Publishing — validating responsiveness, publishing to Tableau Public, and embedding in a Flask web app

3. Data Collection & Extraction

The dataset (Consumption.csv) contains daily electricity usage readings, in Million Units (MU), for Indian states.

Attribute	Detail
Rows (raw)	16,599 daily readings
States covered	33
Regions covered	5 — Northern (NR), Western (WR), Southern (SR), Eastern (ER), North-Eastern (NER)
Date range	02 Jan 2019 – 05 Dec 2020
Fields	State, Region, Latitude, Longitude, Date, Usage (MU)
Missing values	None — the source file arrived pre-cleaned

Note on 2019 vs 2020 comparisons: 2019 has a full 354 unique reporting days in the file, while 2020 has 144, ending in early December. Any year-over-year total should be read as a partial-year comparison for 2020, not a like-for-like annual figure — the dashboard's monthly and average-based views control for this.

4. Storing Data & SQL Operations

The cleaned data was loaded into a MySQL database (electricity_consumption) and processed with SQL to prepare it for Tableau:

- consumption_raw — staging table loaded directly from the CSV via LOAD DATA INFILE
- Duplicate rows (same state + date) removed; null/zero usage rows dropped
- consumption_final — analysis table adding calculated fields: Year, Month, Quarter, Lockdown Period, and City Type (Metro / Non-Metro)
- Indexes added on State, Region, and Year to keep dashboard queries responsive

Full scripts: 01_create_and_load.sql and 02_clean_transform_analyze.sql (attached).

5. Connecting the Database to Tableau

Tableau Desktop connects to the MySQL database as a live or extracted data source (Data > Connect to Server > MySQL). For this project, an extract was used so dashboard interactions stay fast regardless of database load, refreshed whenever the underlying table changes.

6. Data Preparation for Visualization

Even with a clean source file, the following steps were applied before visualization:

- Data review — checked field types, value ranges (Usage: 0.3–522.1 MU), and date coverage per state
- Filtering & structuring — data organized by State, Region, Year, Quarter, and Lockdown Period to match each planned chart
- Field renaming — 'Regions' → Region, 'Dates' → Date, 'Usage' → Usage_MU for clarity in Tableau's field list
- Calculated fields — Quarter, Lockdown Period (Before/After 25-Mar-2020), and City Type (Metro/Non-Metro)
- Validation — aggregated totals cross-checked against source-file sums before publishing

7. Data Visualization

Twelve views were built, each answering a specific question about consumption behavior:

View	What it shows
2019 State Consumption	Total usage per state for 2019, ranked
2020 State Consumption	Total usage per state for 2020, ranked
Total Consumption	All-time total usage per state
Usage By Region	Share of national usage per region
Total Region Consumption	Region totals split by year
Top N / Bottom N	Highest and lowest consuming states
2019 & 2020 Monthwise Consumption	Monthly totals, both years overlaid
Total Consumption Region Wise	Region-level totals
Usage Before/After Lockdown	Total & average daily usage split at 25 Mar 2020
Region-wise State Usage	States within a selected region
Quarter Usage	Q1–Q4 totals by year
Metro City / State Usage	Usage for the six major metro states vs. the rest

Key findings from the data:

- Maharashtra, Gujarat, UP, and Tamil Nadu are consistently the top-consuming states, together accounting for a large share of national demand.
- The Western Region (WR) and Northern Region (NR) lead in total consumption; the North-Eastern Region (NER) consumes the least by a wide margin.
- Sikkim, Mizoram, and Arunachal Pradesh are the lowest consumers, consistent with their smaller populations and industrial base.
- Average daily usage per state was roughly flat comparing the pre- and post-lockdown periods in this dataset, suggesting that any lockdown dip is more visible at a monthly/state level than in the aggregate daily average — worth calling out explicitly in the story rather than assuming a large blanket drop.

8. Dashboard Design

The dashboard combines the individual sheets into a single, filterable view with State, Region, Year, and Date filters, plus Top N/Bottom N controls. Design priorities were:

- User-centered layout — KPIs and the most-asked questions (top states, regional split) surfaced first

- Interactivity — filters let a viewer drill from national → region → state without leaving the page
- Responsiveness — layout adapts from a wide desktop view down to a single-column mobile view

An interactive prototype implementing this same structure (built as a standalone dashboard) is included with this documentation as [electricity_consumption_dashboard.html](#).

9. Data Story

The accompanying Tableau Story walks a viewer through the analysis in sequence:

- Scene 1 — National overview: total consumption and regional split
- Scene 2 — State-level leaders and laggards
- Scene 3 — Seasonal pattern: month-wise and quarter-wise trends
- Scene 4 — The lockdown period: before/after comparison
- Scene 5 — Metro vs. non-metro demand, and closing takeaways

10. Performance Testing

Three factors were checked to keep the dashboard responsive:

- Data volume — table row counts and sizes checked in MySQL Workbench (Table Info panel) before publishing
- Filter impact — State, Region, Year, and Date filters tested individually and combined for query response time
- Visualization complexity — the dashboard was kept to a focused set of sheets to avoid overloading a single view; Top N/Bottom N filters used to cap rendered marks rather than always plotting all 33 states

11. Publishing & Embedding

Publishing steps:

- Finalize the workbook in Tableau Desktop and remove unused sheets/data to reduce file size
- File > Save to Tableau Public, sign in, and name the workbook
- Confirm the published dashboard/story link and set visibility as required

The published dashboard is then embedded into a Flask application via an `<iframe>` pointing at the Tableau Public URL, allowing it to sit inside a custom web UI with any additional authentication the project requires.

12. Conclusion

The analysis confirms that electricity demand in India is concentrated in a small number of industrially and demographically significant states — led by Maharashtra, Gujarat, UP, and Tamil Nadu — and in the Western and Northern regions. The dataset's coverage of the 2020 lockdown period provides a useful, if partial-year, lens on how demand patterns held up during a major disruption. The resulting Tableau dashboard and story make these patterns explorable for a non-technical audience, and the same structure is reproducible end-to-end via the SQL and data-prep scripts included with this documentation.

Appendix: Files Delivered

- `consumption_cleaned.csv` — cleaned dataset with calculated fields (Year, Month, Quarter, Lockdown Period, City Type)
- `01_create_and_load.sql` — MySQL table creation and data load

- 02_clean_transform_analyze.sql — cleaning, calculated fields, and the reference query for every dashboard view
- electricity_consumption_dashboard.html — interactive dashboard prototype (open in any browser)
- video_script.md — narration script for the end-to-end project walkthrough video